Dissecting the Carbon–SLO Tension in Geo-Distributed LLM Serving

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What Does This Paper Do?

A geo-distributed LLM serving simulator, calibrated with A100 / vLLM latency and power measurements.

Schedulers

Four carbon- and SLO-aware policies, from a per-request heuristic to a batch-level facility MILP.

Frontier

The carbon–SLO tradeoff across three models, loads, and carbon-aware and carbon-blind baselines.

Headline result, versus round-robin

Facility MILP reduces operational carbon by **up to 60%** relative to round-robin, with the authors' **average SLO violation of 3%**. The 60% is the upper end. The 3% is an average, not every load.

Takeaway Where a request is served changes both operational carbon and token latency. The lowest-CI site is not automatically the SLO-feasible site.

Source: Topcu et al., Abstract; Sec. I contributions; Sec. VIII. The 26–60% conclusion range is on M20.

Table I: Scope of Prior Work and This Paper

	Formulation		Batch-aware		Overheads		
Study	Placement	Latency target	Latency	Power	Queue	Transfer	Reported latency
SLIT (2025)	Fixed 15-min plan	Min. TTFT, unbounded	×	×	×	✓	Mean TTFT only
Green-LLM (2025)	Fixed 1-h plan	Mean E2E $\leq$ 5 s	×	×	×	✓	None ($ penalty)
FREESH (2025)	Fixed 30-min plan	E2E $\leq 5\times$ profiled	×	×	×	×	E2E viol., p99
GCOS (2026)	Per job (GPU-h)	Job wait $\leq$ 10–20% runtime	–	×	×	✓	Wait violation
MARLIN (2026)	Fixed 15-min plan	Min. TTFT, unbounded	×	×	×	✓	Mean TTFT only
MOER (2026)	Per request	None	×	○	×	×	E2E p95
This paper	Per request and batch	TTFT, TPOT $\leq$ MLPerf	✓	✓	✓	✓	SLO attainment, p50/90/99

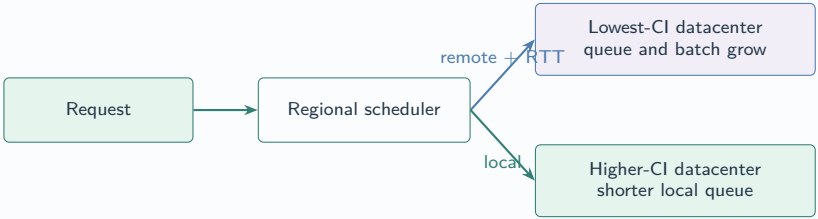
✓ modeled × not modeled ○ partial – not applicable

Broader modeling scope Batch-dependent latency and power, engine queueing, and inter-region transfers under benchmark-based TTFT / TPOT SLOs.

Source: Topcu et al., Sec. I, Table I. Labels abbreviated for readability.

Why Not Always Choose the Greenest Datacenter?

Conceptual schematic, not a measured trace.



Carbon opportunity Carbon intensity varies by more than an order of magnitude across regions and time. Routing toward a lower-CI grid cuts operational carbon.

Latency cost A crowd of requests raises queueing and TTFT, and a larger decode batch raises TPOT. The remote path also pays network RTT.

Takeaway Minimum CI and the SLO-feasible assignment are different decisions once queueing, batching, and RTT are counted.

Source: Topcu et al., Sec. I, paragraphs on spatial scheduling; Sec. II, CI variation. Source-mix medians are on the backup.

What Do Previous Approaches Miss?

Table I compares six carbon-aware geo-scheduling studies with this paper. The gaps are not identical for every row.

Load-independent latency

Most coarse plans (SLIT, Green-LLM, FREESH, MARLIN) treat latency as independent of the live batch. MOER is per request, but it has no latency target.

Queue, KV admission, network

None models engine queueing when KV memory or the batch slot is full. FREESH and MOER also omit inter-region transfer; several others include it.

Phase-dependent power

None fully models prefill versus decode power. MOER is marked partial. The others are unmarked.

Which latency SLO?

Prior targets are mostly self-defined: mean TTFT, a 5 s end-to-end cap, or job wait. This paper uses TTFT and TPOT against MLPerf budgets.

Takeaway The open question is load-dependent TTFT and TPOT under a benchmark SLO, not a new carbon signal by itself.

Source: Topcu et al., Sec. I and Table I. GCOS is a GPU-job scheduler; its latency column is not a token SLO.

What Is the Carbon–SLO Scheduling Problem?

The decision is which datacenter receives the request. It is not a transfer of layers, KV cache, or MoE experts.

$$\mathrm{CO_2} = \sum_{d \in D} E_d \cdot \mathrm{CI}_d, \qquad E_d = \mathrm{PUE}_d \int P_d(t)\, dt$$

P_d sums host power and per-engine power. One PUE and one CI per datacenter.

Operational carbon only

Scope 2 electricity emissions. This paper does not optimize embodied carbon, and it does not trade model quality.

SLO violation

A request violates the SLO if **either** TTFT or TPOT exceeds its budget. Main budgets: 3 s and 100 ms; Mixtral TPOT is 200 ms.

Takeaway Carbon is energy times grid intensity, with PUE. Latency is two separate token budgets, and missing either one counts.

Source: Topcu et al., Sec. II, GHG Protocol and problem formulation; Table II budgets. MLPerf's original TTFT is 2 s; they relax it for geo delay.

How Does the Geo-Distributed Simulator Work?

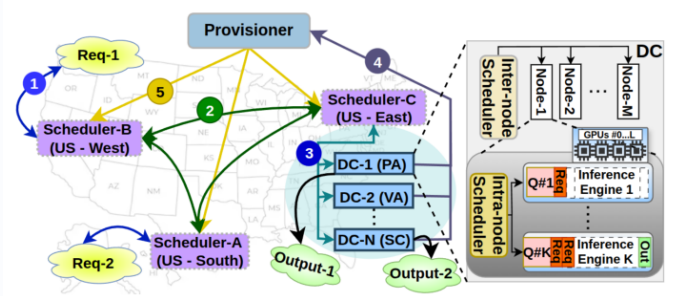


Fig. 3. Overview of the simulator architecture on an example topology.

Figure 3, numbered path

1. Load generator sends a Poisson request to a regional scheduler.

2–3. The scheduler keeps it local or forwards it. A remote hop pays RTT, then the peer scheduler places it.

4–5. The provisioner polls each datacenter and broadcasts the snapshot plus the RTT matrix.

Inside a datacenter, inter-node and intra-node placement are round-robin. The engine runs continuous batching.

Takeaway Scheduling sees a polled per-datacenter snapshot: batch size, queue depth, KV occupancy, prefill duty, and CI.

Source: Topcu et al., Sec. III-C, Fig. 3; Sec. III-A for the polled fields. Savings are attributed to the regional policy because intra-DC placement is round-robin.

How Is LLM Serving Latency Modeled?

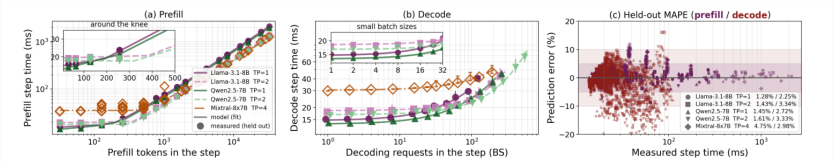


Fig. 1. Per-engine latency model across models and TP degrees: **(a, b)** prefill and decode step time across prefill lengths and batch sizes, respectively; **(c)** prediction error against measured step time, with per-setup MAPE.

Prefill, Fig. 1(a) Nearly flat for short prompts, then roughly linear in prefill tokens. A constant TTFT cannot represent that knee.

Decode, Fig. 1(b) Sublinear in batch size, then sharp after saturation, and higher when the KV context is longer.

Fit, Fig. 1(c) An analytical model calibrated on A100 / vLLM measurements. Held-out MAPE is within 1.7% (dense prefill), 4.8% (Mixtral prefill), and 3.4% (decode).

Takeaway TTFT and TPOT are functions of the live batch, the queue, and the context. They are not constants the scheduler can look up once.

Source: Topcu et al., Sec. III-B, Fig. 1. Coefficients and the exact polynomials are on the backup.

How Are Power and Energy Modeled?

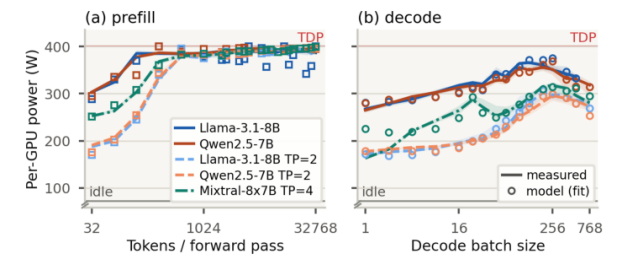


Fig. 2. Per-GPU power during **(a)** prefill and **(b)** decode across models and TP degrees.

Two curves Prefill power reaches TDP once the prompt is long enough. Decode power grows with batch size and stays more sensitive to load.

Energy Each step is average power times step latency. A fast, high-power prefill and a long, lower-power decode do not cost the same.

Always on The power model also charges about 72 W per idle A100 and 300 W of host power. Idle GPUs are not free carbon.

Takeaway Load-dependent energy is part of the carbon price. A scheduler that ignores phase and batch will mis-rank datacenters.

Source: Topcu et al., Sec. III-B, Fig. 2. Held-out step energy is within 5.6% MAPE. TDP shown is 400 W for the A100.

Real GPUs or a Simulator? How Is It Validated?

A. Real hardware

A100-40GB, vLLM v0.6.3, FP16. Latency and power are profiled on this GPU.

B. Analytical fit

Per-phase latency and energy models, one fit per model and tensor-parallel degree.

C. Geo-simulation

Many regions, schedulers, queues, RTT, and carbon. This layer is not another hardware run.

Within-node validation, Llama-3.1-8B only

Three data-parallel vLLM engines, 19 arrival rates, more than 40K requests. TPOT average error below 5%. TTFT average error 9.1%. GPU energy within 2.5% of the measurement.

What this does not certify

The check is single-node inference serving. It is not a validation that 16 physical datacenters were deployed. Mixtral at TP=4 already fills their four-GPU node.

Source: Topcu et al., Sec. V-B; limitations in Sec. VI. TTFT error is 3.5% below saturation, 5.2% in overload, and 28.5% near saturation.

Which Scheduling Policies Are Proposed?

All four share one latency estimator. They differ in how many requests they decide together, and in how they price energy.

Policy	Decision	Load coupling	Energy of a batch
Greedy heuristic	one request	current snapshot only	not shared
Standard ILP	one window, one solve	frozen inside that solve	sum of per-request costs
Iterative ILP	same window, up to 3 solves	re-projected after each solve	still additive
Facility MILP	window, plus a use bit per DC	capacity gated by that bit	wake-up charged once

Greedy α=0 Carbon only. The paper states that this coincides with MOER. α=1 would be latency only. They report 0, 0.5, and 0.9.

Why the ladder Each step is aimed at one failure of the previous step: uncoordinated requests, a stale load estimate, then double-counted batch energy.

Takeaway Facility MILP is not only a larger ILP. It changes who pays the shared energy of a batch.

Source: Topcu et al., Sec. IV opening and Sec. IV-A–C. Solver limits: $T_{\max}$=3, MILP time limit 0.25 s, HiGHS.

How Do Greedy and Standard ILP Work?

Greedy, per request

Keep only SLO-feasible datacenters, then

$$\alpha\,\widehat{\mathrm{Lat}} + (1-\alpha)\,\widehat{\mathrm{CI}}.$$

Latency is TTFT plus OSL times TPOT. Hats divide by the max over feasible datacenters. The request goes to the lowest score.

Weakness: requests in the same polling period herd onto one site. Concurrent schedulers share one snapshot.

Standard ILP, one window

Assign the window jointly:

$$\min \sum c_{r,d} x_{r,d}, \quad \sum_{d} x_{r,d} = 1, \quad \sum_{r} x_{r,d} \le H_d.$$

$c_{r,d} = \mathrm{PUE}_d\, E_{r,d}\, \mathrm{CI}_d$. H_d is remaining batch slots.

The full ILP freezes H_d and the latency estimates for that single solve.

Takeaway Greedy cannot coordinate a window. Standard ILP can, but it still prices every request against the load from before the assignment.

Source: Topcu et al., Sec. IV-A, Eq. (1); Sec. IV-B, Eq. (2). Window: 100 ms or 300 requests. The matrix is totally unimodular.

Why Iterative ILP and Facility MILP?

Iterative ILP

Write the assignment back into each datacenter's batch and queue, recompute TTFT and TPOT, and solve again. Stop on a repeated assignment or after 3 iterations.

Facility MILP

Add a one-time carbon f_d if datacenter d receives any request in the window. Marginal request energy stays in $c_{r,d}$.

$$\min \sum c_{r,d} x_{r,d} + \sum_d f_d y_d, \qquad \sum_r x_{r,d} \le H_d y_d, \qquad y_d \in \{0,1\}$$

An additive model can charge an idle engine's fixed step energy twice when two requests share that batch. The y_d term charges that shared batch energy once.

Takeaway The MILP changes the energy model of a shared batch. It does not power off a datacenter, and it does not remove RTT.

Source: Topcu et al., Sec. IV-B–C. f_d is the carbon of waking d's engines for the window, not a measured facility shutdown.

How Is the System Evaluated?

	Dense	MoE
Models	Llama-3.1-8B, Qwen2.5-7B, TP=1	Mixtral-8x7B, top-2 of 8, TP=4
Token budgets	TTFT 3 s, TPOT 100 ms	TTFT 3 s, TPOT 200 ms
Arrival rate	1200 / 2400 / 4000 req/s	800 / 1200 / 2000 req/s
Engines per DC	32 (4 nodes × 8 GPUs)	8, because each engine uses 4 GPUs

Topology 4 regional schedulers × 4 datacenters = 16 DCs. RTT 85–280 ms. Default PUE 1.1. CI is an Electricity Maps hour, held fixed inside a run. Table II has the full list.

Baselines Carbon-blind: round-robin and least-loaded. Carbon-aware: Green-LLM, and MOER, which matches greedy at α=0.

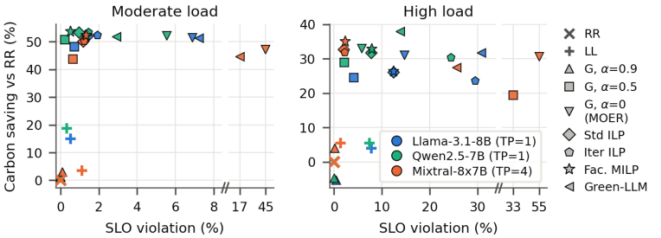
Metrics Operational carbon in gCO_2, including idle and host energy. SLO violation rate. Also TTFT p50 / p90 / p99, because the violation rate hides the tail.

Takeaway The 16 datacenters are simulated roles on one server, not a 16-site hardware deployment. Dense and MoE do not have the same engine count.

Source: Topcu et al., Sec. V-A, Table II. Load labels: low / moderate / high are under 15%, 15–40%, and above 40% of batch capacity under round-robin.

Q1: Can Carbon Be Reduced Without Excessive SLO Violations?

Figure 4(a), Topcu et al. Y: carbon saving versus round-robin. X: fraction of requests missing TTFT or TPOT.



(a) Carbon saving relative to RR and SLO violation for each policy.

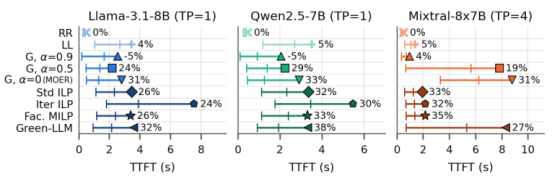
Takeaway Facility MILP keeps most of the carbon saving at a low violation rate. Carbon-only greedy matches the saving only by spending the SLO.

How to read the claim. Up to 60% and the average 3% SLO violation are the authors' summary versus baselines RR, not the coordinate of every marker. Low load is omitted from this figure: facility MILP saves 53.8–59.6% there. The most carbon-aggressive greedy setting reaches up to 55% violation; that is not true of greedy at α=0.9.

Source: Topcu et al., Abstract; Sec. V-C, Fig. 4(a). Moderate-load α=0: up to 6.9% violation on dense models and 44.8% on Mixtral.

Q2: What Happens to Tail Latency Under High Load?

Figure 4(b), Topcu et al., high load only. Bars are TTFT from p50 to p99. The printed percentages are carbon savings versus RR, not latency.



(b) TTFT tail behavior per policy at *high* load, from p50 to p99 with ticks at p50/p90. Labels show carbon savings relative to RR.

Dense models Greedy keeps p99 TTFT inside 3 s, but α=0 still fails the SLO on TPOT: decode batches sit on the green datacenters.

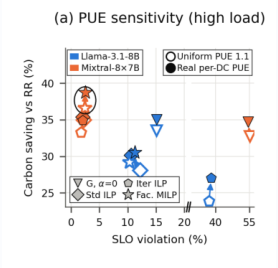
Mixtral Greedy α=0 pushes median TTFT to 3.3 s. ILP and MILP keep the median within 0.7 s and p99 within 2.2 s. Aggressive greedy and Green-LLM reach p99 up to 8.8 s.

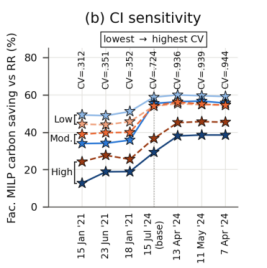
Takeaway A reasonable median is not a tail inside the SLO, and a TTFT tail inside the budget is not a passed SLO violation rate.

Source: Topcu et al., Sec. V-C, Fig. 4(b). Iterative ILP p99 TTFT reaches 7.5 s on the dense models, from repeated solves.

Q3: How Do PUE and Carbon-Intensity Spread Affect Savings?

Figure 5, Topcu et al. (a) high-load PUE. (b) facility MILP across CI hours. Open versus filled markers in (a): uniform PUE 1.1 versus real per-DC PUE.





Real PUE is 1.08–1.32. Facility MILP moves from 29.2% to 30.5% (Llama) and 36.6% to 38.7% (Mixtral). SLO barely changes.

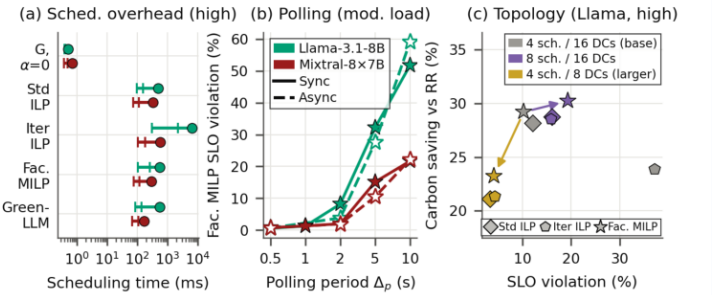
From the lowest CV (0.31) to the highest (0.94), savings rise and the SLO stays flat. High load still saves up to 39–45% when the spread is large.

Takeaway A wider PUE and CI spread creates the carbon opportunity. Higher load spends it. PUE mostly rescales carbon, not the SLO.

Source: Topcu et al., Sec. V-C, Fig. 5. The 13% Llama point at high load and the lowest CV is outside the conclusion's 26–60% summary.

Q4: How Important Are Polling, Overhead, and Topology?

Figure 6, Topcu et al. (a) scheduler CPU time at high load. (b) facility-MILP SLO violation at moderate load. (c) Llama, high load.



(a) Solver ≠ GPU
Greedy is under 1 ms. Facility MILP p99 is up to 0.5 s. Iterative ILP p99 is 6.5 s on Llama.

(b) Polling period
At most 2 s: violation stays at most 8.2%. At 5 s it rises to 10.5–32%.

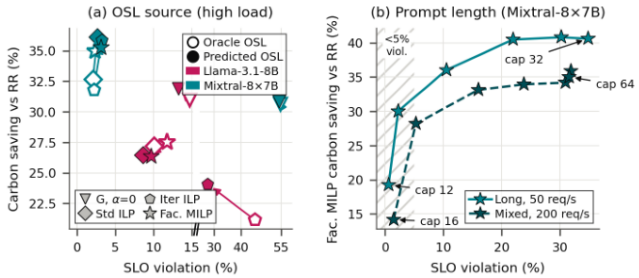
(c) Topology
Eight schedulers: violation 10.3% to 19.4%. Fewer, larger DCs: violation 4.1%, saving 23.2%.

Takeaway Token SLO tracks how fresh the load snapshot is. The milliseconds and seconds in panel (a) are scheduler time, not TTFT.

Source: Topcu et al., Sec. V-C, Fig. 6. Asynchronous polling does not consistently beat synchronous polling.

Q5: What If Output Length Is Unknown?

Figure 7, Topcu et al. (a) high load, oracle versus predicted OSL. (b) Mixtral prompt length as batch capacity grows.



Why OSL is in the score

Decode time and decode energy scale with output length. The paper's example: OSL 100 occupies an engine about 10× longer than OSL 10.

Predicted OSL

Savings move by at most 1.3 points on Llama and 4.1 on Mixtral. Violations rise by at most 1.1 points, except iterative ILP on Llama.

Longer prompts

Near a 5% violation, facility MILP still saves up to 30% (long) and 28% (mixed).

Takeaway These policies tolerate a rough output-length estimate. The carbon–SLO result does not require the oracle length.

Source: Topcu et al., Sec. IV and Sec. V-C, Fig. 7. Exact-bucket accuracy is only 51% (Llama) and 37% (Mixtral); adjacent buckets cover 99% and 70%.

Conclusion

1. Problem

Chasing a lower-CI datacenter moves TTFT and TPOT through queueing, batching, and RTT.

2. Contribution

A within-node-calibrated simulator and four schedulers. Facility MILP prices shared batch energy instead of summing isolated requests.

3. Evidence

Versus round-robin: up to 60% less operational carbon, average SLO violation 3%. Savings grow with CI spread, shrink with load, and depend on snapshot freshness.

4. Limitations, and what is not this paper

Not done here: multi-turn KV locality and heterogeneous hardware. Our direction — heterogeneous GPUs, HBM/HBF, attention and expert placement, cross-DC carbon-aware serving — is not an author result.

Source: Topcu et al., Sec. VI and Sec. VIII. The conclusion's 26–60% band is across loads at the base setting; Fig. 5(b) leaves that band when the CI hour changes.

Discussion Index

Main is the talk. This index opens the evidence behind a question.

Topic	What is here	Jump
Gap	Table I, row by row, including partial cells	B01
Definitions	CI, PUE, Scope 2, TTFT, TPOT, violation	B02
Step models	Latency and energy formulas, MAPE, idle power	B03
Validation	19 rates, error by regime, single-node limit	B04
Engine	Continuous batching, KV admission, queueing	B05
Greedy / ILP	Equation (1), Equation (2), frozen load	B06
MILP	Iteration, wake-up bit, what y_d does not do	B07
Setup	Table II, baselines, load labels, not 16 real sites	B08
Sensitivity	PUE, CV, polling, topology, OSL, prompt length	B09
Limits	Reproducibility and claims we would not make	B10

Simulator trust M10 B04 **Main result** M15 M16 **Method** M13 B07

Go to Conclusion (M20)

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Table I: What Each Prior Scheduler Models

Check marks follow Table I: modeled, not modeled, partial, or not applicable. "Ours" is this paper.

Study	Placement and latency target	Lat. / power / queue / transfer	Reported
SLIT	15-min plan; min TTFT, unbounded	no / no / no / yes	mean TTFT
Green-LLM	1-hour plan; mean E2E ≤ 5 s	no / no / no / yes	none ($ penalty)
FREESH	30-min plan; E2E $\leq 5\times$ profiled	no / no / no / no	E2E viol., p99
GCOS	per GPU-job; wait $\leq$ 10–20% runtime	n/a / no / no / yes	wait violation
MARLIN	15-min plan; min TTFT, unbounded	no / no / no / yes	mean TTFT
MOER	per request; no latency target	no / partial / no / no	E2E p95
Ours	per request and batch; TTFT and TPOT at MLPerf	yes / yes / yes / yes	SLO, p50/90/99

Do not flatten the column "Most" of the coarse planners assume load-independent latency. That sentence in Sec. I cites SLIT, Green-LLM, FREESH, and MARLIN. It is not a claim that MOER or GCOS make the same latency assumption.

Transfer is mixed Queueing from KV admission is unmarked for every prior row. Inter-region transfer is marked for SLIT, Green-LLM, GCOS, and MARLIN, and unmarked for FREESH and MOER.

Source: Topcu et al., Sec. I, Table I.

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Discussion Index

Carbon Accounting, TTFT, TPOT, and the SLO

Carbon intensity Generation-weighted: $\mathrm{CI} = \sum_g (P_g / P_{\mathrm{total}}) \cdot \mathrm{CI}_g$. The paper quotes median CI below 50 $\mathrm{gCO_2/kWh}$ for nuclear and solar, 490 for natural gas, and 820 for coal. CI varies by more than an order of magnitude across regions and time.

Scope 2 Operational emissions over intervals are $\sum_k E_{\mathrm{tot}}[k] \cdot \mathrm{CI}[k]$, with $E_{\mathrm{tot}} = P_{\mathrm{tot}} \Delta t$ and $P_{\mathrm{tot}} = P_{\mathrm{IT}} \times \mathrm{PUE}$. Across datacenters, $\mathrm{CO_2} = \sum_d E_d \cdot \mathrm{CI}_d$.

Inside one datacenter $E_d = \mathrm{PUE}_d \int P_d(t)\, dt$ and P_d adds host power to every engine's power. One PUE and one CI per datacenter, because the site shares the grid and the facility.

TTFT Queueing plus prefill, plus scheduling delay and inter-region RTT when the placement is remote.

TPOT Average decode-step time, including steps stalled by another request's prefill.

SLO violation The fraction of requests that miss either the TTFT budget or the TPOT budget. Main study: 3000 ms / 100 ms. Mixtral TPOT is 200 ms. Long and mixed prompts use a 4500 ms TTFT. MLPerf's single-node TTFT is 2 s; only TTFT is relaxed.

Not this paper's objective No embodied-carbon term, no SSD amortization, and no quality-for-carbon trade.

Source: Topcu et al., Sec. II; Table II; Sec. III-A.

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Analytical Latency and Power Formulas

Prefill time, for total tokens N, batch B, and prompt lengths n_i:

$$t_{\mathrm{pre}} = t_{\mathrm{tok}}(N) + b_0 + b_1 B + b_2 \sum_i n_i^2.$$

t_{tok} is piecewise linear in N. Its slope increases after a saturation point k.
Decode time is quadratic in batch size. Each coefficient is linear in the mean cached length $\ell = K/B$:

$$t_{\mathrm{dec}} = a_0 + a_1 B + a_2 B^2, \qquad a_j = \alpha_j + \beta_j \ell.$$

Step energy, not power alone:

$$E_{\mathrm{pre}} = e_0^{\mathrm{pre}} + e_p N + e_a \sum_i n_i^2, \qquad E_{\mathrm{dec}} = e_0^{\mathrm{dec}} + e_d B + e_k K.$$

MAPE Prefill step time: within 1.7% for Llama-3.1-8B and Qwen2.5-7B at both TP degrees, and 4.8% for Mixtral-8x7B at TP=4. Decode step time: within 3.4% on all five setups. Held-out step energy: within 5.6%.

Standing power Idle engine: about 72 W per A100. Host: 300 W, of which the CPU package is about 146 W under load on their testbed. Calibration: vLLM v0.6.3, FP16, A100, prompts from 32 to 32K tokens.

Why decode MAPE is the weaker fit Decode steps are tens of milliseconds, so thermal and frequency transients show up. The paper states that limit.

Source: Topcu et al., Sec. III-B, Figs. 1–2. Symbols follow the paper's prose; the PDF does not number these displays as equations.

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Within-Node Validation, and Its Boundary

What was replayed Three data-parallel Llama-3.1-8B vLLM engines, TP=1, A100-40GB, CNN-DailyMail, round-robin, Poisson rates from 2 to 50 req/s. The simulator then replays each logged engine, arrival time, input length, and actual output length.

Where those rates sit Measured saturated capacity is 27.58 req/s. The sweep covers 7% to 181% of that capacity, including the queue-free regime and overload.

Errors, 19 rates, more than 40K requests TPOT moves from 15.5 ms to 54 ms as prefills stall decode; average error is below 5%. TTFT moves from 71 ms to more than 20 s; overall average error is 9.1%. Split: 3.5% below saturation, 5.2% in overload, 28.5% around saturation, where the simulator overestimates TTFT.

Independence and energy One engine instead of three, at the same per-engine load, changes TPOT by 3.3%. Simulated GPU energy stays within 2.5% of the measurement at every rate.

Boundary Sec. VI: validation is within a node and for Llama-3.1-8B only. A single Mixtral-8x7B engine at TP=4 already fills the four-GPU node, so Mixtral has no matching multi-engine end-to-end check. Absolute carbon on other GPUs would move. The 16-datacenter experiments remain simulation.

Source: Topcu et al., Sec. V-B and Sec. VI. Step-model MAPE in Sec. III-B is a separate check from this end-to-end replay.

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Continuous Batching, KV Capacity, and Queueing

Admission Each step retires finished requests, then admits queued ones while the batch limit and the new KV both fit. Queueing delay depends on occupancy. It is not a fixed constant.

Engine budgets Batch size follows vLLM `max_num_seqs`; the per-step token budget follows `max_num_batched_tokens`. A long prefill is split across steps and stalls the running decode batch.

KV example Llama-3.1-8B uses 128 KiB of KV per token after the weights. Batch capacity is 40 at TP=1, 80 at TP=2, and 400 for Mixtral.

TTFT estimate Scheduling delay, RTT if remote, wait for a slot, and prefill:
$\mathrm{TTFT}_{r,d} = \mathrm{RTT}(s, s') + W_d + t_{\mathrm{pre}}(\mathrm{ISL}_r + Q_d)$.

TPOT estimate $\mathrm{TPOT}_{r,d} = t_{\mathrm{dec}}(B_d{+}1, \mathrm{KV}_d) + T_d^{\mathrm{pre}}/(1 - \mathrm{duty}_d)$. The $+1$ is this request. Duty stretches decode by prefills already running.

Per-datacenter snapshot The provisioner polls batch size, queue tokens, KV occupancy, prefill duty, and CI. The paper uses averages so the poll and the decision stay small.

Source: Topcu et al., Sec. III-C and the latency estimator in Sec. IV. Intra-DC schedulers are round-robin.

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Greedy Score and the Standard ILP

Equation (1). $\mathcal{F}_r$ is the set of datacenters whose estimated TTFT and TPOT meet the SLO. $\mathrm{Lat}_{r,d} = \mathrm{TTFT}_{r,d} + \mathrm{OSL}_r \cdot \mathrm{TPOT}_{r,d}$.

$$\mathrm{score}_{r,d} = \alpha \frac{\mathrm{Lat}_{r,d}}{\max_{d' \in \mathcal{F}_r} \mathrm{Lat}_{r,d'}} + (1-\alpha) \frac{\mathrm{CI}_d}{\max_{d' \in \mathcal{F}_r} \mathrm{CI}_{d'}}.$$

α=0 is carbon only and matches MOER. α=1 is latency only. Reported settings are 0, 0.5, and 0.9. Herding: every request in a polling period can select the same site, and regional schedulers act on one shared snapshot.

Equation (2). $x_{r,d} \in \{0,1\}$, $c_{r,d} = \mathrm{PUE}_d \cdot E_{r,d} \cdot \mathrm{CI}_d$, and $H_d = \max(0, \mathrm{cap}_d - \mathrm{load}_d)$.

$$\min_x \sum_{r \in R} \sum_{d \in \mathcal{F}_r} c_{r,d} x_{r,d} \quad \text{s.t.} \quad \sum_{d \in \mathcal{F}_r} x_{r,d} = 1, \quad \sum_{r \in R} x_{r,d} \le H_d.$$

The constraint matrix is totally unimodular, so the LP relaxation is integral. HiGHS solves it in tens of milliseconds. H_d and the latency estimates stay fixed during this one solve: a datacenter that is feasible for each request alone can miss the SLO after the requests land together.

Fallback If the solver fails or cannot place a request inside capacity, the scheduler uses the lowest-CI SLO-feasible datacenter with headroom, which is greedy at α=0, and round-robin if none exists.

OSL Both TPOT's queued-prefill term and the carbon estimate need the output length. The main experiments use the oracle length. The predicted length is only the Fig. 7 study.

Source: Topcu et al., Sec. IV-A–B, Eqs. (1)–(2); Sec. V-A fallback and OSL.

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Iterative ILP and the Facility MILP

What iteration updates After solve t, $n_d^{(t)} = \sum_r x_{r,d}^{(t)}$ is projected onto that datacenter's batch size and queue as if the window had already been dispatched. TTFT, TPOT, the feasible set, and $E_{r,d}$ are recomputed. Equation (2) is solved again until two successive assignments match or $T_{\max} = 3$.

What iteration does not change Carbon is still a sum of per-request costs. Capacity and latency constraints couple the requests. The energy model does not.

Wake-up bit The facility MILP adds $\sum_d f_d y_d$ to the objective and replaces the capacity row with $\sum_r x_{r,d} \le H_d y_d$, $y_d \in \{0,1\}$. $y_d = 1$ means d receives at least one request from the window. f_d is the carbon of waking d's engines. Two requests on an idle engine then pay that fixed step energy once, not twice.

Size of the extra model Only $|D|$ new binaries, against $|R| \cdot |D|$ assignment variables. The paper says HiGHS still finishes inside the 0.25 s limit.

Does not power off This is an accounting device for shared batch energy inside the window. It does not power off a datacenter building, it does not delete inter-region RTT, and idle engines still remain in the later discussion of engine activation.

Source: Topcu et al., Sec. IV-B–C. The engine-activator remark in Sec. VI is an observation, not a claim that the MILP already implements it.

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Table II, Baselines, and Load Labels

Item	Value in the paper
Regions	4 schedulers: EU-North, US-East, Asia, South America. 4 DCs each. EU-North examples: Netherlands, Belgium, Lithuania, Latvia.
Hardware roles	4 nodes per DC, 8× A100-40GB per node. Simulated. The sweep runs on one 64-core server, 200 s per policy-rate pair.
Network and PUE	Inter-region RTT 85–280 ms, AWS one-year p50. Default PUE 1.1. Real per-DC PUE is only the Fig. 5(a) sweep.
CI	Electricity Maps, hourly, held constant within a run.
Load labels	Mean round-robin batch / batch capacity: low <15%, moderate 15–40%, high >40%. Rates 1200/2400/4000 (7B/8B) and 800/1200/2000 (Mixtral).
Models	Llama-3.1-8B and Qwen2.5-7B are dense, TP=1. Mixtral-8x7B is MoE, 13B active, TP=4, so a quarter as many engines per DC.
Scheduler knobs	Poll 1 s (swept 0.5–10 s). Window 100 ms or 300 requests. Greedy $\alpha \in \{0, 0.5, 0.9\}$. Oracle OSL except Fig. 7.

Round-robin and least-loaded Carbon-blind, one request at a time. Intra-DC placement is also round-robin, so measured savings are credited to the regional scheduler.

Green-LLM and MOER Green-LLM is ported as scheduling decisions only. MOER sends each request to the lowest-CI SLO-feasible datacenter, which is greedy at α=0.

Not a 16-site deployment No claim in the PDF says these 16 datacenters were physically built for the experiment.

Source: Topcu et al., Sec. V-A, Table II.

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Figure 5–7: Values Behind the Charts

Panels are on Main 17–19. Savings below are versus round-robin. Sec. V-C.

PUE, high load Reported PUE 1.08 (Netherlands) to 1.32 (Japan). Facility MILP: Llama 29.2% to 30.5%, Mixtral 36.6% to 38.7%. Japan in this note is CI 449–502 with PUE 1.32. SLO barely moves. These baselines are not the 26% / 35% labels on Fig. 4(b); the PDF does not reconcile them.

CI spread 35K hours, 2021–2024. CV 0.31 to 0.94. Llama/Mixtral: low 49/44% to 59/55%; moderate 34/39% to 56/55%; high 13/24% to 39/45%. SLO flat. The 13% point is outside the 26–60% conclusion sentence.

Overhead, polling Greedy <1 ms. Facility MILP p99 up to 0.5 s. Iterative ILP p99 6.5 s on Llama. Poll ≤ 2 s: violation at most 8.2%. At 5 s: 10.5–32%. Async is not consistently better.

Topology Llama, high load, facility MILP. Doubling schedulers: violation 10.3% to 19.4%, savings almost unchanged. Fewer, larger DCs: violation 4.1%, saving 23.2%. Engine count is matched.

Predicted OSL MLP, 128 units, about 27K parameters, <0.1 ms. Exact bucket 51% Llama and 37% Mixtral (random guess 17%). Adjacent included: 99% and 70%. Savings move ≤ 1.3 and ≤ 4.1 points. Violations ≤ 1.1 points, except iterative ILP on Llama.

GovReport and mixed prompts Mean ISL 10.7K at 50 req/s, and 2.9K at 200 req/s. TTFT budget 4.5 s. Facility MILP saves up to 30% at batch capacity 16 (long) and 28% at 24 (mixed), near 5% violation.

Source: Topcu et al., Sec. V-C, Figs. 5–7. Fig. 4(b) labels and the PUE paragraph are both quoted; they are not averaged together.

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Limits, Reproducibility, and Open Questions

What the PDF shows A calibrated simulator and a scheduling comparison under MLPerf-derived token SLOs. The carbon result is versus round-robin, stated as an average and as a frontier, not as one guaranteed point.

Validation Within a node, Llama-3.1-8B, three engines. Not a 16-datacenter hardware deployment. Other GPUs would change absolute latency and carbon. The authors expect the queueing pattern to carry over, and they do not show that.

Workload Single-turn requests. Long prompts stress KV capacity. Multi-turn sessions, where prefix KV ties the next turn to one datacenter, are future work in the paper.

Not an extra system Facility MILP leaves Llama TP=1 engines idle 73% / 56% / 33% of the time at low / moderate / high load, versus at most 10% for round-robin. Idle plus host energy is 28–81% of remaining carbon. The engine activator is only motivated. At TP=2, facility MILP emits 22% more carbon at high load.

Our direction Heterogeneous GPUs, HBM/HBF, attention and expert placement, and cross-datacenter carbon-aware serving. Our direction is not an author result.

Reproducibility Code: `github.com/topcuburak/carbon-slo-at-geoscale`. This deck does not re-run it. If the repository and the PDF disagree, the PDF remains the source of these claims.

Source: Topcu et al., Sec. VI and Sec. VIII. Claude assistance is disclosed by the authors in the acknowledgment; they say the technical claims are theirs.

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